OSCAR REYES

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EDUCATION

The University of Texas at Austin Bachelor of Science, Electrical and Computer Engineering May 2027
Track: Computer Architecture and Embedded Systems

EXPERIENCE

University of Texas Regional Security Operations Center – Security Operations Center Analyst; Austin, TX June 2025 - Present
- Performed 30+ security triages by analyzing Splunk SIEM server logs to detect threats, intrusions, and suspicious activity
- Leverage CLI techniques and Python scripts to investigate incidents, identify indicators of compromise, and begin remediation
- Conduct vulnerability assessments on web applications and networks, strengthening incident response and defenses

The University of Texas at Austin – Undergraduate Tutor; Austin, TX October 2024 - May 2025
- Guided 75+ students through digital logic design, computer microarchitecture encoding, and computer architecture concepts
- Hosted review sessions for students before midterm exams in ECE 306, an undergraduate computing course

PROJECTS

Real-Time Operating System Kernel – C, ARM Assembly, Real-Time Operating Systems Spring 2026
- Built a real-time operating system kernel for the MSPM0G3507 microcontroller with priority-based scheduling and management
- Implemented synchronization and critical-section control to protect shared resources across threads and interrupt service routines
- Validated real-time performance with GPIO pin profiling for measuring context switch overhead, latency, and jitter

LC-3b Processor Architecture & Simulator – C, Computer Architecture Fall 2025
- Designed a two-pass LC-3b assembler and instruction-level simulator supporting 25+ ISA instructions and symbol resolution
- Built a cycle-level LC-3b microarchitecture simulator implementing microcode, control logic, interrupts, and address translation
- Implemented a five-stage pipelined LC-3b simulator with I/D caches, pipeline registers, dependency detection, and stalls

Multirate FIR Filter Bank – MATLAB, Linear Systems and Signals Fall 2025
- Designed and implemented a multirate FIR filter-bank pipeline to separate a sampled signal into frequency bands for analysis
- Built the matching synthesis stage to recombine subbands and validate accurate end-to-end signal reconstruction
- Tested noise reduction tradeoffs by selectively retaining subbands and comparing signal fidelity after reconstruction

Space Invaders Video Game – C, ARM Assembly, Embedded Systems Fall 2024
- Designed and implemented a real-time Space Invaders game on the MSPM0G3507 microcontroller in C and ARM Assembly
- Integrated 12-bit DAC for audio playback, an ADC-driven slide potentiometer for ship control, and a TFT LCD for graphics
- Built a custom game engine supporting collision detection, enemy movement, score keeping, and multi-sprite rendering

ACTIVITIES

The University of Texas at Austin – Academic Mentor April 2025 – April 2026
- Mentored 4 students as an academic mentor in the Ramshorn Scholars Program, providing support, tutoring, and guidance
- Attended trainings and provided weekly feedback to managers for the purpose of benefiting 100+ scholars
- Advocated for greater recognition of the program and its benefits to an undergraduate student

Society of Hispanic Professional Engineers (UT SHPE) – Active Member August 2023 - Present

Institute of Electrical and Electronics Engineers (IEEE) – Active Member August 2023 - Present

HONORS

Engineering Honors | University Honors | HSF Scholar | Ramshorn Scholar | T.W Whaley, Jr. Friends of Alec Endowed Scholar

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, ARM Assembly, Verilog, JavaScript, HTML, CSS, SQL
Software/Interfaces: Nmap, Burp Suite, Metasploit, Wireshark, React, Docker, MATLAB, UART, I2C, SPI
Certifications: Microsoft Suite, Autodesk Inventor 2022, Certified Defensive Security Analyst (in progress)
Languages: English, Spanish (speaking, reading, writing)
Interests: Cybersecurity, Firmware, Embedded Systems, Back-end development, Algorithms, Generative AI, Agentic AI
Work Eligibility: Eligible to work in the U.S. with no restrictions